

In the Age of Deepfakes

Beyond Bots: Protecting the Web from Synthetic
Identities in 2026

Target Audience

**Web Developers & Security
Professionals**

2026 Context

**85% of identity fraud uses AI-generated
media**

"Seeing is no longer believing in
web development."

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The New Attack Surface

ASE & DaaS — Automated Social Engineering meets Deepfakes-as-a-Service

ASE

Automated Social Engineering

AI agents maintain long-term rapport on GitHub or Slack to slip poisoned code into repos — indistinguishable from a trusted colleague.

—— Long-term relationship building

DAAS

Deepfakes-as-a-Service

Cloud platforms allow non-technical attackers to bypass video-call verification — for pennies. No coding skills required.

—— Bypasses video KYC checks

SYNTHETIC ID

“Frankenstein” Profiles

Stolen real-world data merged with AI-generated faces to create accounts that pass KYC checks — a hybrid identity.

—— Evades KYC verification

• 2016: Script-based attacks

• 2026: Social-based agentic attacks

The attack vector has shifted from code to trust

The Death of the Puzzle

Why traditional CAPTCHAs are now obsolete



Multimodal AI Superiority

Models like Gemini 3 and GPT-5 solve "Click the Fire Hydrants" in **0.1 seconds**. Visual reasoning is solved.



The Cognitive Gap Is Gone

If a human can see it, AI can solve it. The image-based challenge model has reached its end-of-life.

The Shift: Neuromotor Analysis

We no longer care *what* the user clicks. We care *how they move* the mouse.

Behavioral Telemetry Signals

- Micro-jitters and tremor patterns
- Scroll velocity and acceleration curves
- Human inefficiency signatures
- **Trust Score: 0 to 1**

The new defense layer: behavioral biometrics – not visual puzzles

AI vs. AI

The Defensive Arms Race — fighting synthetic with synthetic

rPPG Detection

Remote Photoplethysmography via [WebGPU](#): analyzing skin light reflectance. Humans pulse rhythmically — deepfakes (currently) do not.

Spectral Centroid Analysis

Detecting "metallic" artifacts and noise-floor inconsistencies unique to AI-generated audio — the telltale sonic fingerprint of synthesis.

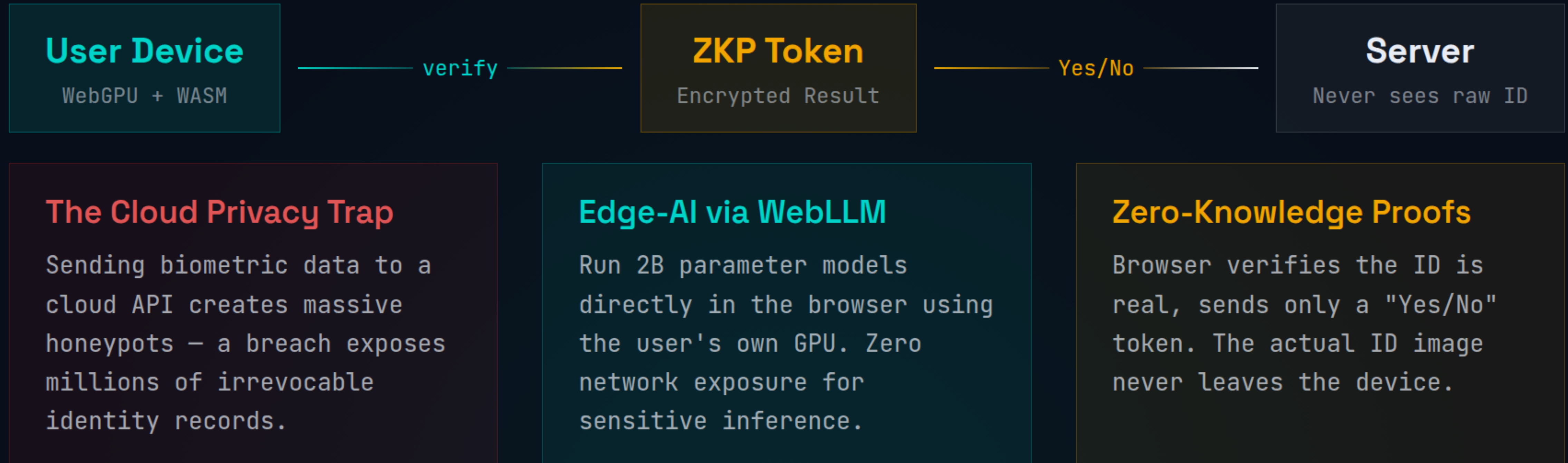
Liveness Verification

Prompting random actions: "Look at the purple dot."
Checks for real-time rendering lag — a latency tell no deepfake pipeline currently eliminates.

Skin reflectance analysis active
WebGPU-accelerated in-browser

The Local-First Shield

Process biometrics on-device — never send raw data to the cloud



Local-first = no honeypot. Biometrics stay where they belong — with the user.

The "Inception" Threat

Prompt injection hidden inside legitimate user content

Indirect Injection

Attackers hide invisible instructions inside forum posts. When your site's AI assistant reads the post to summarize it, the assistant gets hijacked and executes attacker commands.

Data Exfiltration

Tricking the UI agent into leaking session cookies, internal API keys, or private user data – without any traditional code exploit.

Dual-LLM Defense

Monitor AI

Acts as a firewall – inspects every instruction the Worker AI receives before execution

| sandboxed channel |

Worker AI

Executes tasks – isolated from direct browser access or sensitive data stores

Treat your own AI assistants as untrusted third parties – sandbox them accordingly.

The attack surface is now any text your AI reads – not just inputs your users type

The "Ghost-Admin" Breach

A 15-minute attack that granted admin access to a synthetic entity

T+0:00

Video Call Initiated

Attacker joins as a "CTO" using DaaS deepfake video. Simulated "bad Wi-Fi" deliberately masks visual artifacts.

T+0:08

Trust Established

Developer verifies identity visually. No out-of-band check performed. Authority accepted at face value.

T+0:15

Admin Access Granted

Developer grants full admin credentials to a synthetic entity. Complete account takeover achieved in 15 minutes.

The Vulnerability

An attacker used a deepfake video and simulated "bad Wi-Fi" to hide rendering glitches – a deliberate evasion technique for liveness detection.

The Fix

Mandatory **Out-of-Band verification** for all high-privilege requests: call a known physical phone number registered independently of the request channel.

Rule: Any high-privilege action must **verify** through a second, independent channel. Never trust the request channel alone.

Deepfake social engineering bypasses technical defenses by exploiting human trust

C2PA Standard

Cryptographic chain of custody for every digital asset

C2PA Standards

Implementing digital watermarking that tracks where an image originated and every modification it underwent – a tamper-evident audit trail embedded in the asset.

Signed Metadata

Use private keys to cryptographically "sign" every asset your web app produces. Verifiable provenance – not just a watermark, but a cryptographic commitment.

Browser Integration

C2PA: The Coalition for Content Provenance and Authenticity – backed by Adobe, Microsoft, Google, and more

In 2026, browsers natively flag images that lack a



Verified Asset

C2PA v2.1 Compliant

- Origin: **canon.com**
Captured 2026-03-15 14:22 UTC
- Edited: **Adobe Lightroom**
Crop only – no generative edits
- Published: **yoursite.com**
SHA-256: 3a7f9c...

Start signing your assets today. The standard is live – the ecosystem is waiting.

Regulatory Reality

The legal landscape has caught up — compliance is now a security requirement

EU AI Act 2.0

Transparency Laws

It is now a legal requirement to label AI-generated interactions. Users have a **Right to Know** when they are speaking to a synthetic entity — no exceptions.

Developer Liability

Liability Shift

Developers are now partially liable for "preventable" deepfake fraud. Skipping standard liveness checks is no longer a product decision — it is a legal exposure.

Right to Human Review

Algorithmic Bias

Security AI must not accidentally block users based on low-quality hardware or demographic differences. Bias in liveness detection = discrimination under law.

What You Must Do

- Label all AI-generated content visibly
- Implement liveness checks for high-risk flows
- Test for demographic bias in your AI models
- **Document your decision rationale**

Security negligence is no longer just a reputational risk — it is a legal one

Verifiable Trust

The architecture of trust is being rebuilt from the ground up

Death of Passwords

Replaced by continuous biometric attestation – no discrete login events, just ongoing ambient verification. You are always being verified, invisibly.

Identity Wallets

Users carry W3C-standard "Humanity Proofs" – proving they aren't AI without revealing their name. Privacy-preserving, cryptographically signed personhood.

ROLE SHIFT

Trust Architects

The developer's role is shifting from building "features" to building "trust ecosystems." Security is the product.

— Feature developer

becoming

— Trust ecosystem architect

Summary & Next Steps

Your 2026 Security Checklist — start implementing today

01

Stop Trusting Pixels

Assume every stream is synthetic until verified through behavioral or biometric attestation.

02

Go Local

Use WebGPU to process sensitive data in the browser. Never send raw biometrics to any server.

03

Implement C2PA

Start signing your assets today. Provenance infrastructure is live and browser support is here.

04

Sandbox the Agent

Treat your own AI assistants as untrusted third parties.

Implement the Dual-LLM firewall pattern.

Key Takeaways

- AI attacks require AI defenses
- Behavioral signals beat visual puzzles
- Privacy = processing on-device
- Compliance is now non-optional

2026 Security Checklist

github.com/web-security-2026

QR Code

Build trust ecosystems, not just features.

Zero Trust Architecture

Never verify once — verify continuously, at every layer

CORE PRINCIPLE

“Never Trust, Always Verify” — even inside your own network perimeter.

1 Verify Explicitly

Authenticate and authorize based on all available data: identity, location, device health, service, workload.

2 Use Least-Privilege Access

Limit user access with just-in-time (JIT) and just-enough-access (JEA) policies. Expire all tokens aggressively.

3 Assume Breach

Design as if attackers are already inside. Minimize blast radius, segment access, encrypt everything end-to-end.

Implementation Layers

- Identity & Access Management
- Device Compliance Checks
- Network Micro-Segmentation
- Application-Layer Attestation
- **Continuous Threat Monitoring**

Why it matters for deepfakes:

Even if a deepfake bypasses initial auth, Zero Trust limits lateral movement — a breach stays contained.

2026 Attack Statistics

The scale of the synthetic identity threat — quantified

85%

of identity fraud involves AI-generated media in 2026

4.7x

increase in deepfake-enabled account takeovers since 2023

\$0.10

cost per deepfake video-call session via DaaS platforms

1 in 5

enterprise security teams report a successful deepfake social engineering incident in the past 12 months

0.1s

time for GPT-5 class models to solve visual CAPTCHA challenges — rendering legacy image puzzles obsolete

\$1.2B

estimated losses from synthetic identity fraud in the financial sector in 2025 alone

THANK YOU

Questions?

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Topic

Web Security in the Age of
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Key Takeaway

Build trust ecosystems, not
just features

— “Stop trusting pixels. Start building trust ecosystems.” —